

23030301曹磊第四次作业

1.考虑下面的表格结构文法 G2 :

$$S \rightarrow a | \wedge | (T)$$

$$T \rightarrow T, S | S$$

(1)给出 $(a, (a, a))$ 和 $((a, a), \wedge, (a)), a)$ 的最左和最右推导。

字符串 1: $(a, (a, a))$

最左推导:

$$\begin{aligned} S &\Rightarrow (T) \\ &\Rightarrow (T, S) \\ &\Rightarrow (S, S) && (T \rightarrow S) \\ &\Rightarrow (a, S) && (S \rightarrow a) \\ &\Rightarrow (a, (T)) && (S \rightarrow (T)) \\ &\Rightarrow (a, (T, S)) && (T \rightarrow T, S) \\ &\Rightarrow (a, (S, S)) && (T \rightarrow S) \\ &\Rightarrow (a, (a, S)) && (S \rightarrow a) \\ &\Rightarrow (a, (a, a)) && (S \rightarrow a) \end{aligned}$$

最右推导:

$$\begin{aligned} S &\Rightarrow (T) \\ &\Rightarrow (T, S) \\ &\Rightarrow (T, (T)) && (S \rightarrow (T)) \\ &\Rightarrow (T, (T, S)) && (T \rightarrow T, S) \\ &\Rightarrow (T, (T, a)) && (S \rightarrow a) \\ &\Rightarrow (T, (S, a)) && (T \rightarrow S) \\ &\Rightarrow (T, (a, a)) && (S \rightarrow a) \\ &\Rightarrow (S, (a, a)) && (T \rightarrow S) \\ &\Rightarrow (a, (a, a)) && (S \rightarrow a) \end{aligned}$$

字符串2: $((a, a), \wedge, (a)), a)$ 的推导

最左推导:

$$\begin{array}{ll}
S \Rightarrow (T) & \\
\Rightarrow (T, S) & (T \rightarrow T, S) \\
\Rightarrow (T, a) & (S \rightarrow a) \\
\Rightarrow (S, a) & (T \rightarrow S) \\
\Rightarrow ((T), a) & (S \rightarrow (T)) \\
\Rightarrow ((T, S), a) & (T \rightarrow T, S) \\
\Rightarrow ((T, (T)), a) & (S \rightarrow (T)) \\
\Rightarrow ((T, (S)), a) & (T \rightarrow S) \\
\Rightarrow ((T, (a)), a) & (S \rightarrow a) \\
\Rightarrow ((T, S), a) & (S \rightarrow (T)) \\
\Rightarrow (S, a) & (T \rightarrow S) \\
\Rightarrow (((T), S), a) & (T \rightarrow S \rightarrow (T)) \\
\Rightarrow (((T, S), S), a) & (T \rightarrow T, S) \\
\Rightarrow (((T, S), \Lambda), a) & (S \rightarrow \Lambda) \\
\Rightarrow (((T, S), \Lambda), a) & \\
\Rightarrow (((T, (T)), \Lambda), a) & (S \rightarrow (T)) \\
\Rightarrow (((T, (S)), \Lambda), a) & (T \rightarrow S) \\
\Rightarrow (((T, (a)), \Lambda), a) & (S \rightarrow a) \\
\Rightarrow (((T, S), \Lambda), a) & (S \rightarrow (T)) \\
\Rightarrow (((S, S), \Lambda), a) & (T \rightarrow S) \\
\Rightarrow (((a, S), \Lambda), a) & (S \rightarrow a) \\
\Rightarrow (((a, a), \Lambda), a) & (S \rightarrow a)
\end{array}$$

最右推导：

$$\begin{array}{ll}
S \Rightarrow (T) & \\
\Rightarrow (T, S) & \\
\Rightarrow (T, a) & (S \rightarrow a) \\
\Rightarrow (T, S) & (S \rightarrow a) \\
\Rightarrow ((T), S) & (T \rightarrow S \rightarrow (T)) \\
\Rightarrow ((T, S), S) & (T \rightarrow T, S) \\
\Rightarrow ((T, (T)), S) & (S \rightarrow (T)) \\
\Rightarrow ((T, (S)), S) & (T \rightarrow S) \\
\Rightarrow ((T, (a)), S) & (S \rightarrow a) \\
\Rightarrow ((T, S), S) & (S \rightarrow (T)) \\
\Rightarrow ((T, S), a) & (S \rightarrow a) \\
\Rightarrow ((T, S), a) & \\
\Rightarrow (((T), S), a) & \\
\Rightarrow (((T, S), S), a) & \\
\Rightarrow (((T, (T)), S), a) & \\
\Rightarrow (((T, (S)), S), a) & \\
\Rightarrow (((T, (a)), S), a) & \\
\Rightarrow (((T, S), S), a) & \\
\Rightarrow (((S, S), S), a) & \\
\Rightarrow (((a, S), S), a) & \\
\Rightarrow (((a, a), S), a) & \\
\Rightarrow (((a, a), \Lambda), a) & \\
\Rightarrow (((a, a), \Lambda, (a)), a) &
\end{array}$$

(2) 指出 $((a, a), \Lambda, (a)), a$ 的规范规约及每一步的句柄。根据这个规范规约，给出“移进-规约”的过程，并给出它的语法树自下而上的构造过程。

规范规约及句柄（最右推导的逆）

规范规约的句柄是当前句型中最左边的一个短语，它使得对该短语进行规约后，剩余部分仍能被最右推导出来。

步骤	句型	规约动作	句柄 (α)	产生式 ($\alpha \rightarrow A$)
1	$((a, a), \Lambda, (a)), a$	规约	a	$S \rightarrow a$
2	$((a, a), \Lambda, (a)), S$	规约	a	$S \rightarrow a$
3	$((a, a), \Lambda, (S)), S$	规约	S	$T \rightarrow S$
4	$((a, a), \Lambda, (T)), S$	规约	(T)	$S \rightarrow (T)$
5	$((a, a), \Lambda, S), S$	规约	Λ	$S \rightarrow \Lambda$
6	$((a, a), S, S), S$	规约	a	$S \rightarrow a$
7	$((a, S), S, S), S$	规约	a	$S \rightarrow a$
8	$((S, S), S, S), S$	规约	S, S	$T \rightarrow T, S$
9	$((T), S, S), S$	规约	(T)	$S \rightarrow (T)$
10	$((S, S, S), S)$	规约	S, S	$T \rightarrow T, S$
11	$((T), S)$	规约	(T)	$S \rightarrow (T)$
12	(S, S)	规约	S, S	$T \rightarrow T, S$
13	(T)	规约	(T)	$S \rightarrow (T)$
14	S	接受		

步骤	符号栈	输入串	规约动作 (句柄 $\rightarrow$ 非终结符)
0	#	$((a, a), \Lambda, (a)), a \#$	预备
1	#(	$((a, a), \Lambda, (a)), a \#$	移进
2	#((	$(a, a), \Lambda, (a)), a \#$	移进
3	#(((	$a, a), \Lambda, (a)), a \#$	移进
4	#(((a	$, a), \Lambda, (a)), a \#$	移进
5	#(((S	$, a), \Lambda, (a)), a \#$	规约: $S \rightarrow a$

步骤	符号栈	输入串	规约动作 (句柄 $\rightarrow$ 非终结符)
6	#(((T	, a), Λ , (a)), a)#	规约: $T \rightarrow S$
7	#\$((T, \$	a), Λ , (a)), a)#	移进
8	#(((T, a	), Λ , (a)), a)#	移进
9	#(((T, S	), Λ , (a)), a)#	规约: $S \rightarrow a$
10	#(((T	), Λ , (a)), a)#	规约: $T \rightarrow T, S$
11	#(((T)	, Λ , (a)), a)#	移进
12	#((S	, Λ , (a)), a)#	规约: $S \rightarrow (T)$
13	#((T	, Λ , (a)), a)#	规约: $T \rightarrow S$
14	#\$((T, \$	Λ , (a)), a)#	移进
15	#((T, Λ	, (a)), a)#	移进
16	#((T, S	, (a)), a)#	规约: $S \rightarrow \Lambda$
17	#((T	, (a)), a)#	规约: $T \rightarrow T, S$
18	#\$((T, \$	(a)), a)#	移进
19	#((T, (	a)), a)#	移进
20	#((T, (a	)), a)#	移进
21	#((T, (S	)), a)#	规约: $S \rightarrow a$
22	#((T, (T	)), a)#	规约: $T \rightarrow S$
23	#((T, (T))	, a)#	移进
24	#((T, S	, a)#	规约: $S \rightarrow (T)$
25	#((T	, a)#	规约: $T \rightarrow T, S$
26	#((T)	, a)#	移进
27	#(S	, a)#	规约: $S \rightarrow (T)$
28	#(T	, a)#	规约: $T \rightarrow S$
29	#\$(T, \$	a)#	移进
30	#(T, a	)#	移进
31	#(T, S	)#	规约: $S \rightarrow a$
32	#(T	)#	规约: $T \rightarrow T, S$
33	#(T)	#	移进
34	#S	#	规约: $S \rightarrow (T)$
35	#S	#	接受

$$3. T \rightarrow S \implies LASTVT(S) \subseteq LASTVT(T)$$

$$4. T \rightarrow T, S \implies LASTVT(S) \subseteq LASTVT(T)$$

$$\implies \mathbf{LASTVT(T) = \{a,), , \}}$$

(4) 计算G2的优先关系表，G2是一个算符优先文法吗？

	a	Λ	(	)	,	#
<i>a</i>				>	>	>
Λ				>	>	>
(	<	<	<	≡	<	
)	>	>		>	>	>
,	<	<	<	>	<	
#	<	<	<			≡

对于G2的任何终结对(a,b),至多只有一种关系成立,所以 G2是一个算符优先文法